A function f can be thought as a mechanism (or device) (Fig.1.12(b)), which gives a unique output $f(x)$ to every input x.

DO YOU KNOW?

A function is also called as a mapping or transformation.

Note

If $f : X \to Y$ is a function then

- The set X is called the domain of the function f and the set Y is called its co-domain.
- If $f(a) = b$, then b is called 'image' of a under f and a is called a 'pre-image' of b.
- The set of all images of the elements of X under f is called the 'range' of f.
- $f : X \to Y$ is a function only if
  - (i) every element in the domain of f has an image.
  - (ii) the image is unique.
- If A and B are finite sets such that $n(A) = p$, $n(B) = q$ then the total number of functions that exist from A to B is q^p.
- In this chapter we always consider f to be a real valued function.
- Describing domain of a function

(i) Let $f(x) = \dfrac{1}{x+1}$. If $x = -1$ then $f(-1)$ is not defined. Hence f is defined for all real numbers except at $x = -1$. So, domain of f is $\mathbb{R} - \{-1\}$.

(ii) Let $f(x) = \dfrac{1}{x^2 - 5x + 6}$; If $x = 2, 3$ then $f(2)$ and $f(3)$ are not defined. Hence f is defined for all real numbers except at $x = 2$ and 3. So, domain of $f = \mathbb{R} - \{2, 3\}$.

Progress Check

1. Relations are subsets of _____. Functions are subsets of _____.
2. True or False: All the elements of a relation should have images.
3. True or False: All the elements of a function should have images.
4. True or False: If $\mathrm{R} : A \to B$ is a relation then the domain of $\mathrm{R} = A$.
5. If $f : \mathbb{N} \to \mathbb{N}$ is defined as $f(x) = x^2$ the image of 1 and 2 are _____ and _____.
6. What is the difference between relation and function?
7. Let A and B be two non-empty finite sets. Then which one among the following two collection is large?
   - (i) The number of relations between A and B.
   - (ii) The number of functions between A and B.

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